

AutoRefund AI — Technical Documentation

Production-Ready AI Customer Support Agent with Deterministic Policy Enforcement

1. Problem Statement

In modern e-commerce automation, traditional LLM-based customer support bots suffer from a critical flaw: FINANCIAL HALLUCINATION. Standard AI models may promise refunds to customers without verifying backend database records, customer order ownership, delivery dates, or product return eligibility rules.

Key Challenges Solved:

- Unreliable LLM Decision Making: Pure text generation models cannot be trusted with autonomous monetary transactions.
- Lack of Real Backend Integration: Chatbots often operate on isolated text without dynamic CRM database validation.
- Policy Compliance Risks: Store return windows (e.g., 30-day limits) and non-refundable categories (Digital Software, Gift Cards, Perishable items) are easily bypassed by raw generative AI.
- Security Vulnerabilities: Trusting frontend client requests for customer identity or refund amounts enables fraudulent overrides.

2. Project Motivation

The primary motivation of AutoRefund AI is to demonstrate how to architect a production-ready AI Agent that combines the natural language understanding of LLMs with the absolute mathematical reliability of server-side code.

By enforcing a strict boundary between AI Tool Orchestration and Deterministic Policy Validation, the application guarantees 100% compliance with business rules while delivering a smooth, empathetic customer experience.

3. Core Objectives

1. Server-Side Policy Authority: Ensure the LLM NEVER acts as the final decision maker for money approvals.
2. Dynamic Tool Orchestration: Implement an iterative agent loop that dynamically calls backend tools (getCustomer, getOrder, validateRefundEligibility, processRefundDecision).
3. Deterministic Database CRM Integration: Seed exactly 15 realistic customer profiles with complete order histories, delivery dates, and product categories.
4. Real-Time Telemetry & Auditing: Provide an Admin Operations Dashboard displaying structured step-by-step agent execution logs and metrics.
5. Fail-Safe Reliability: Include an automated deterministic fallback runner so tests and live demos function 100% reliably with or without external API keys.

4. Technology Stack & Knowledge Used

- **Framework:** Next.js 15+ (App Router, React 19, TypeScript)
- **Styling:** Vanilla Tailwind CSS + Glassmorphism design tokens
- **Database & ORM:** Prisma ORM with SQLite (dev.db) & PostgreSQL readiness
- **AI Orchestration:** OpenAI Function Calling API (gpt-4o-mini) + Fallback Loop
- **Validation:** Zod schema validation for API payloads and tool parameters
- **Testing:** Vitest automated test suite (16 business logic tests passing)
- **Logging:** Structured database event logging (AgentLog model)
- **Deployment:** Vercel-compatible serverless architecture

5. System Architecture & Flow

User Request (Next.js Chat UI)

```
% % % > Next.js API Route (/api/chat) [Zod Validation]
  % % % > OpenAI AI Agent Loop (Tool Selection)
    % % % > getCustomer(query)
    % % % > getOrder(orderId)
    % % % > validateRefundEligibility(orderId, customerId)
    %   % % % > Deterministic Policy Engine (Server-Side Code)
      %       % % % 30-Day Delivery Window Check
      %       % % % Order Ownership Verification
      %       % % % Product Category Check
      %       % % % Previous Refund Check
      %       % % % Refund Amount Cap Check
    % % % > processRefundDecision(orderId, customerId, decision)
      % % % Updates CRM Order Status to REFUNDED
      % % % Creates RefundRequest record
      % % % Records Structured Agent Execution Logs
```

6. Pre-Seeded CRM Scenarios

The CRM database contains 15 realistic customer profiles covering all required edge cases:

- Case 1: Valid Refund (Sarah Jenkins / ORD-1001) -> APPROVED (\$149.99 Electronics)
- Case 2: Window Exceeded (Mark Davis / ORD-1002) -> DENIED (Delivered 45 days ago)
- Case 3: Already Refunded (Elena Rostova / ORD-1003) -> DENIED (Status REFUNDED)
- Case 4: Non-Refundable Item (David Chen / ORD-1004) -> DENIED (Digital Download Software)
- Case 5: Ownership Mismatch (Amanda Taylor querying ORD-1001) -> DENIED (Not Owner)
- Case 6: Multi-Order Customer (James Wilson / ORD-1006A) -> APPROVED
- Cases 7-15: Custom Engraved items, In-Transit orders, Gift Cards, Ergonomic Office Chairs, Bluetooth Speakers, and Perishable Coffee Beans.